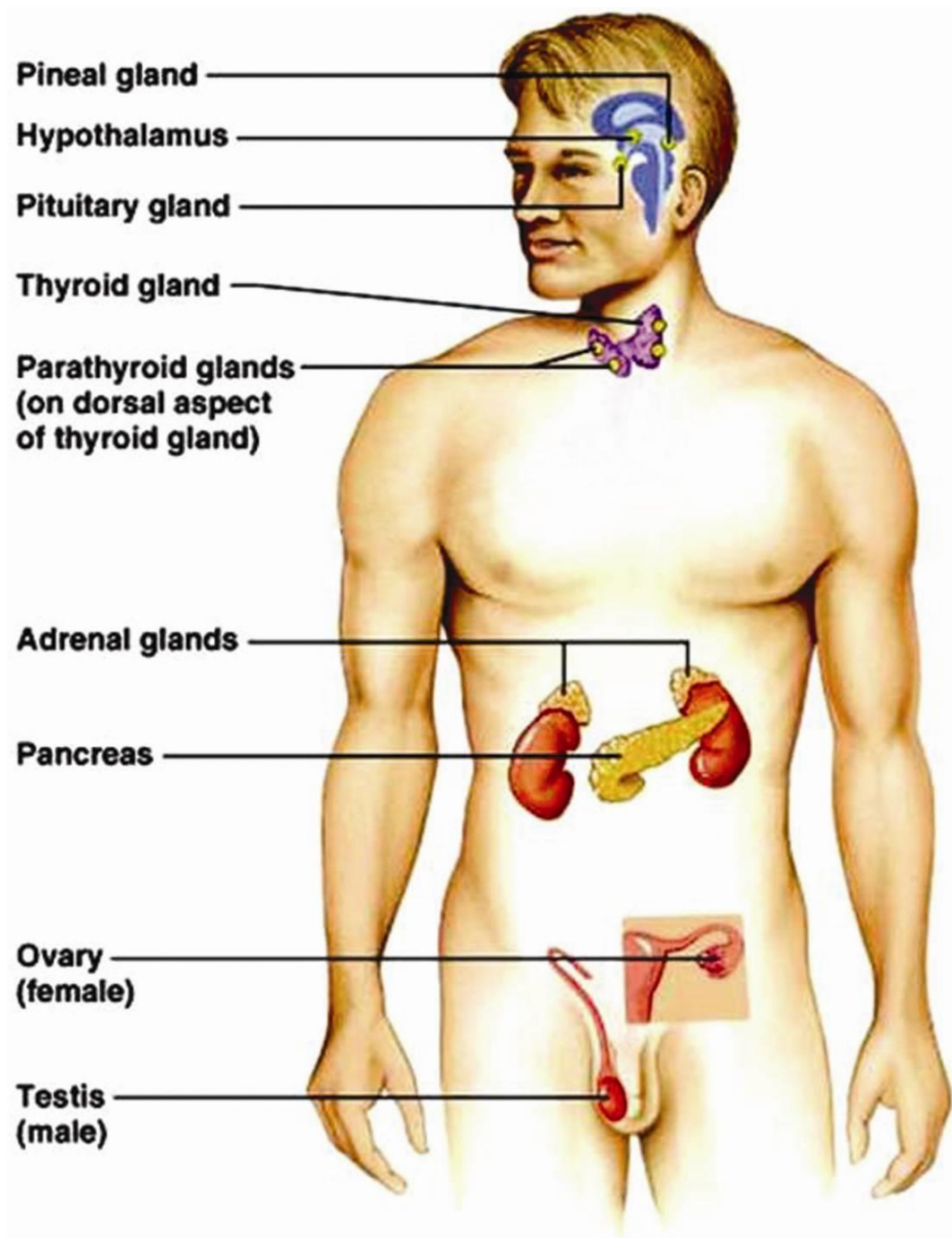


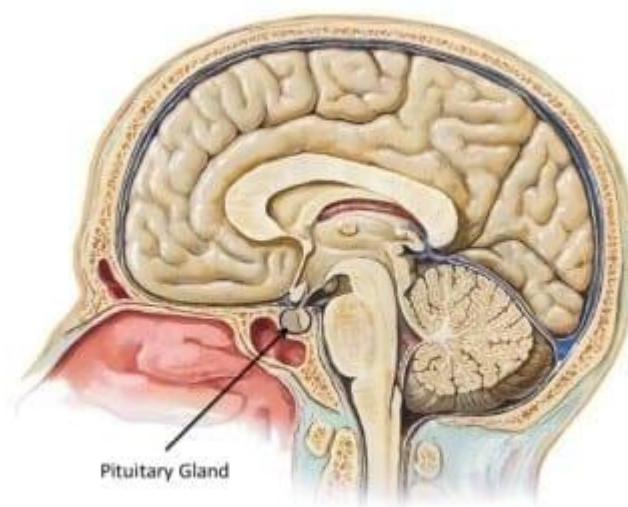
ENDOCRINE SYSTEM



★ A group of **ductless** glands that **secrete** hormones directly into the **blood stream**. They are:

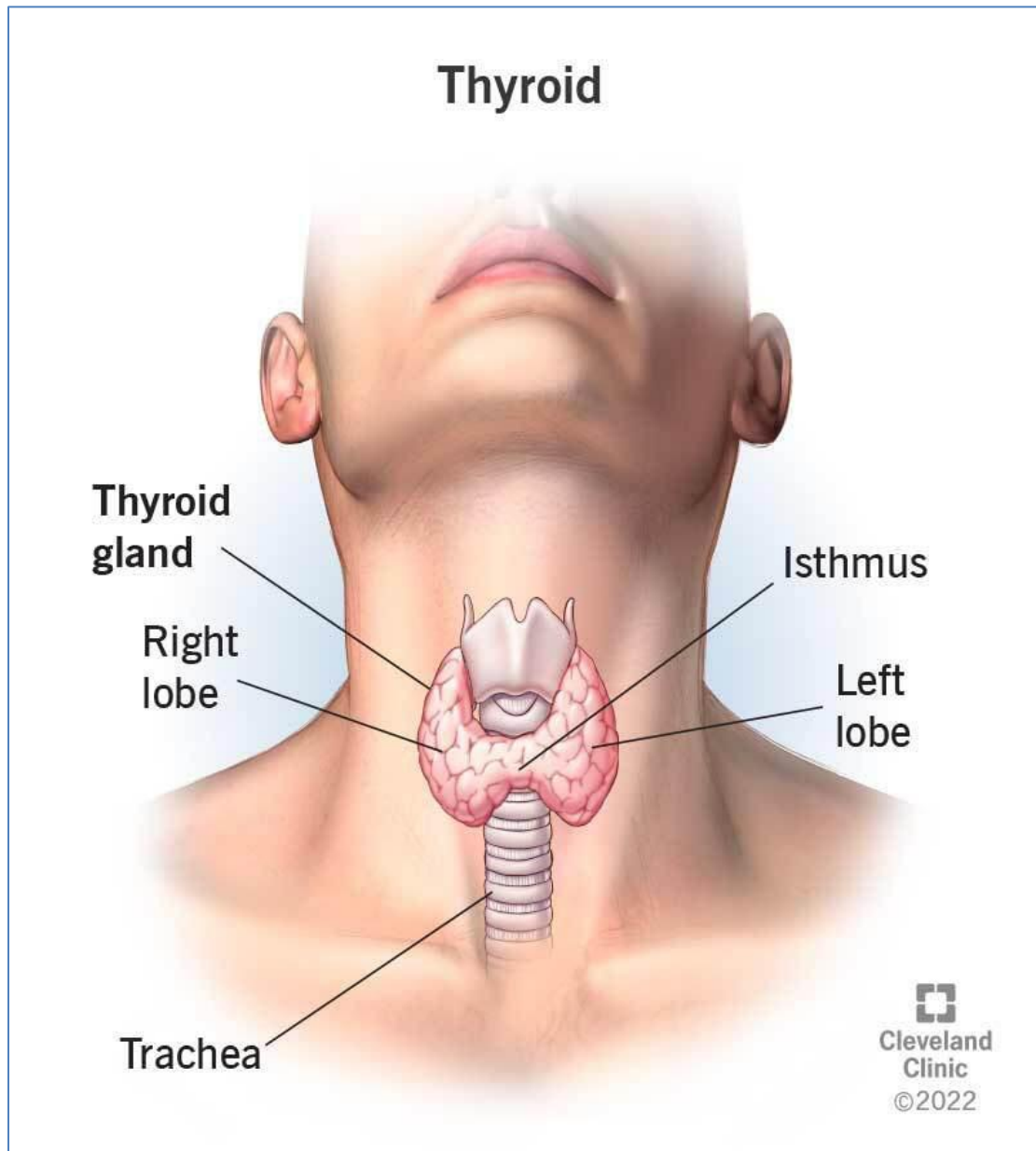
1- Pituitary gland

- A small gland 1/2 g in weight which lies in a special fossa in the base of the skull, below the brain.
- It is formed of **anterior and posterior lobes**.
- It secretes **many hormones** which circulate in the blood and **control the secretion of all the remaining endocrine glands**, as TSH, ACTH, prolactine hormone and growth hormone.



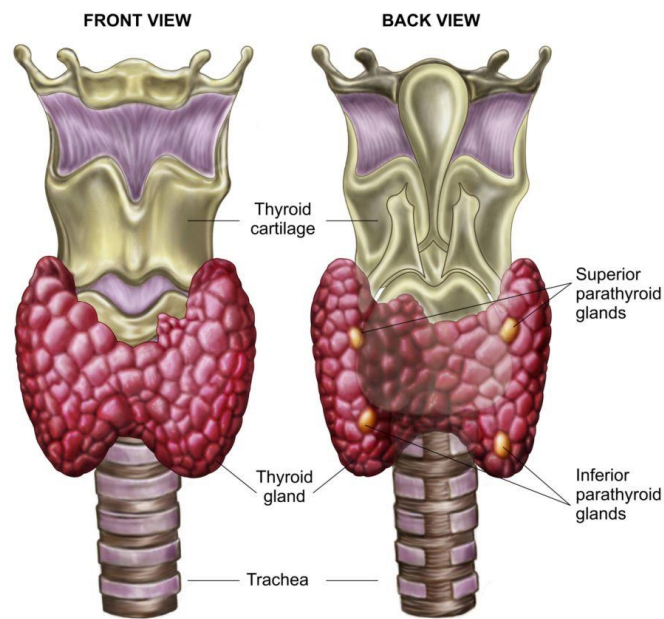
2- Thyroid gland

- **Butterfly-shaped** gland which lies in the **lower part of front of the neck, surrounding the trachea**.
- It consists of **2 pyramidal-shaped lobes** which are connected together by a transverse part called **isthmus**.
- It **moves** up and down with **swallowing**.
- It secretes **thyroxin hormone**.
- Enlargement of thyroid gland called **goiter**.



3- Parathyroid glands

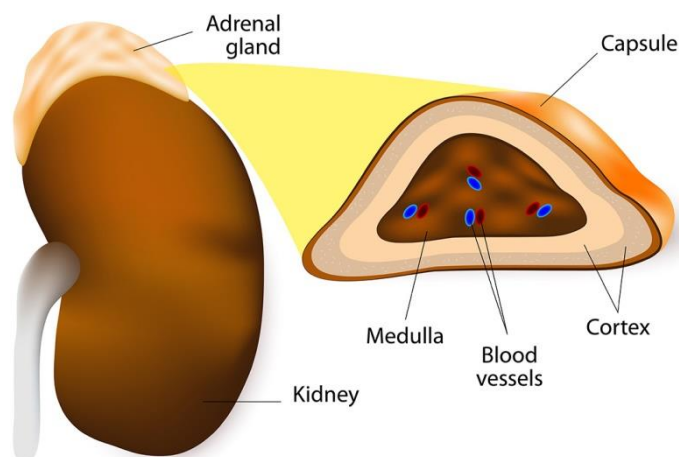
- These are **4** small yellowish glands embedded in the **posterior surface of the thyroid gland**
- They are responsible for secretion of **parathyroid hormone** which regulates the **calcium and phosphorus metabolism**.



4- Suprarenal glands (Adrenals)

- Two **semilunar or triangular** glands present at the **upper poles of the kidneys**.
- Each gland has **cortex and medulla**.
- The **cortex** secretes **cortisone and aldosterone** hormones while the **medulla** secretes **adrenaline and noradrenaline** hormones.

ADRENAL GLAND



5- Pancreas

- A **mixed** endocrine and exocrine gland which lies obliquely on the **posterior abdominal wall**, extending from the duodenum till the spleen.
- It is **consists of** head (in the C-shaped duodenum), neck, body and tail.
- Its endocrine part **secretates** insulin and glucagon hormones that regulate the blood glucose level.

6- Gonads

- a- **2 Testis** in male secretes testosterone hormone.
- b- **2 Ovaries** in female secretes estrogen and progesterone hormones.

Lymphatic system

- **Functions:** It is a system which is responsible for:
 1. Drainage of high molecular weight waste products from the tissues to the large veins.
 2. Providing the main immune function of the body.
 3. Absorption and transport of fat from the small **intestine**.
- **Anatomy of lymphatic system:** It is formed by:

A) Aggregation of cells called **lymphocytes** in specific lymphatic organs; as lymph nodes (عقد ليمفاوية), spleen (طحال), tonsils and thymus gland.

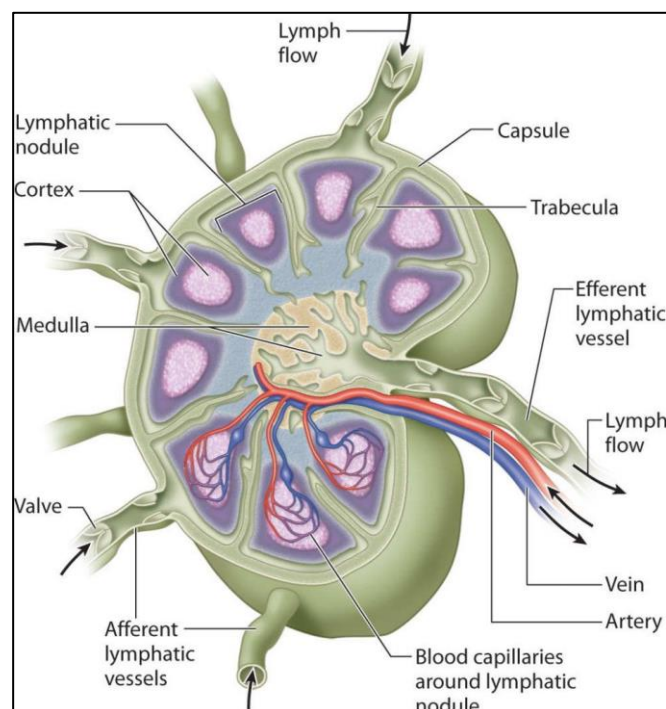
B) Lymphatic channels which consists of the following:

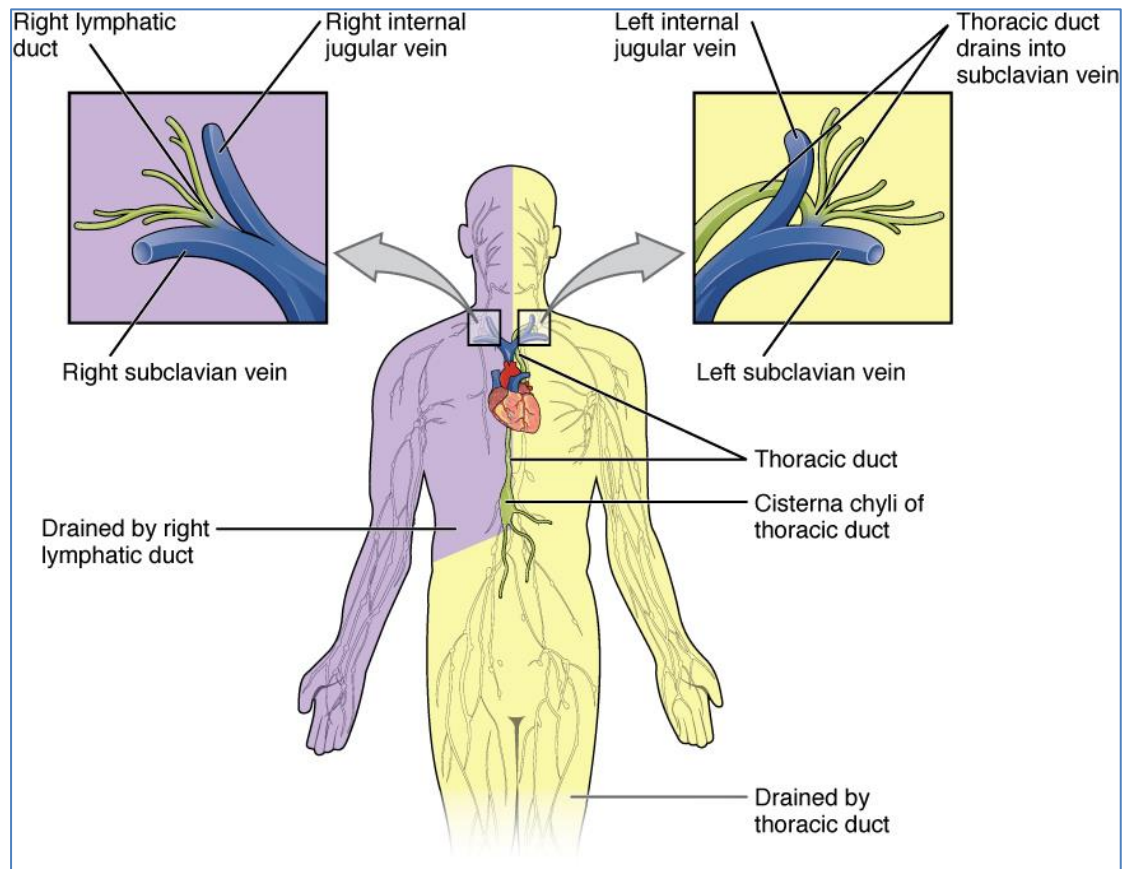
Lymphatic capillaries ... small diameter
Lymphatic vessels larger diameter
Lymphatic trunks larger diameter
Lymphatic ducts larger diameter

C) Lymph: It is the fluid which passes through the lymphatic channels.

- * The fluid circulating inside lymph vessels is called **lymph** which is rich in lymphocytes that combat infection, and contain large particles in the tissue fluid.
- * Lymph fluid formation starts in the **intercellular tissue spaces** and flows inside the lymph vessels.
- * The lymph fluid drains finally into the **venous blood**.
- * The lymphatic vessels are interrupted at various sites by masses of lymphoid tissues called **lymph nodes**.
- * The lymph vessels have valves.
- * The **thoracic duct** is the largest and longest lymphatic vessel which drains the whole body except the upper right quadrant of the body which drains into **right lymph trunk**. These lymph vessels drain into **large veins** in the root of the neck.
- * Lymph vessels are **absent** in the CNS, cornea, epidermis of skin and cartilage.

LYMPH NODE





- ✓ The violet area is drained by right lymphatic duct.
- ✓ The yellow area is drained by thoracic duct.

• **Applied anatomy of lymphatic system**

Lymphadenitis

It is inflammation of lymph nodes in case of infection.

Lymphangitis

It is inflammation of lymphatic vessels in case of infection.

Lymphadenopathy

- ✓ It is enlargement of the lymph nodes.
- ✓ It may be generalized (affecting all groups of lymph nodes) or localized (affecting one group of lymph nodes).
- ✓ It occurs due to infection or malignancy.

